



# Visa Token Service

Implementation Guide for Participant-Branded  
Solutions (Host Card Emulation—Issuer Wallet)

Visa Supplemental Requirements

Version 1.4.4



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## Introduction

The *Visa Token Service Implementation Guide for Participant-Branded Solutions (Host Card Emulation—Issuer Wallet)* provides a detailed description of implementation steps applicable to an issuer as well as the implementation activities required to integrate with Visa Token Service (VTS) using an issuer-developed Mobile Application on an Android™ or iOS mobile device that supports Visa payWave for near-field communication (NFC) using Host Card Emulation (HCE). This document is intended to be read in conjunction with the supporting documents listed in Table 1, Related Publications.

## Audience

This document is for a project implementation team (including business, technology development, and project management) to follow the processes required for effective management of a VTS issuer wallet integration project.

The business considerations and implementation information in this document are directed to issuer staff that will manage the implementation, activation, and ongoing support of a tokenization program. This document is intended for issuers that plan to use VTS for the generation, provisioning, and lifecycle management of payment tokens provisioned to an issuer's own Mobile Application.

**Note:** This document only applies to payment tokens provisioned to an issuer's own Mobile Application. For information about non-issuer (i.e., third party) Mobile Applications, issuers should refer to the *Visa Token Service Implementation Guide for Issuer Participation in VDEP Third-Party Solutions*.

## Scope

This document, along with the documents referenced in Table 1, provides the necessary information to enable an issuer to integrate with VTS.

VTS is intended to address the enablement of multiple digital payment use cases over time. This version of the document addresses Visa payWave for Android and iOS mobile devices that support NFC using HCE on issuer-developed Mobile Applications.

## Related Publications

Table 1 provides a list of documentation that issuers and solution providers can use to obtain additional technical details, system processing requirements, and implementation information for the initial use cases supported by VTS.

**Table 1: Related Publications**

| Title                                                                                                                       | Description                                                                                                                                                                                                                                                                                 |
|-----------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>EMV® Payment Tokenisation Specification Technical Framework</i>                                                          | Describes the payment token system landscape, the types of entities that provide key support for the use of payment tokens, the details to implement multiple use cases, and the benefits of adopting a unified approach. Available from <a href="http://www.emvco.com">www.emvco.com</a> . |
| <i>Security Requirements and Evaluation Guidance for Mobile Applications – Visa Digital Solutions</i>                       | Outlines Visa’s testing guidance to Visa labs testing Mobile Applications for compliance against Visa’s Mobile Application security requirements.                                                                                                                                           |
| <i>Visa Card Metadata Management (VCMM) User Guide</i>                                                                      | VCMM is an online tool to help manage card art configuration, terms and conditions, and other metadata at account range level for tokenized BINs. This guide is intended to assist clients to successfully manage card art and metadata configuration for the BINs enabled for VTS.         |
| <i>Visa Cloud-Based Payments Program Description</i>                                                                        | Provides a high-level overview of the Visa cloud-based payments program. Describes how cloud-based payments work and the benefits and features for issuers and solution providers.                                                                                                          |
| <i>Visa Cloud-Based Payments Program Minimum Requirements and Guidelines</i>                                                | Provides minimum requirements and guidelines for implementation of Visa cloud-based payments.                                                                                                                                                                                               |
| <i>Visa Contactless Payment Specification Version 2.0.2, including Additions and Clarifications 3.0 (or higher version)</i> | Provides the specification for the global interoperable contactless solution.                                                                                                                                                                                                               |
| <i>Visa payWave for Mobile Issuer Implementation Guide</i>                                                                  | Provides issuers with information required to implement a Visa payWave for mobile program.                                                                                                                                                                                                  |
| <i>Visa Ready Program for Cloud-Based Payments Program Guide</i>                                                            | Provides a description of the compliance and testing programs for cloud-based payments administered by the Visa Ready Program office.                                                                                                                                                       |

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| Title                                                                                                 | Description                                                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Visa Risk Manager (VRM) User Guide: Rules Manager</i>                                              | VRM is a risk decision engine offered to clients as part of Visa Token Service (VTS) to manage rules related to provisioning decisions. This guide assists clients in creating and publishing rules related to VTS. |
| <i>Visa Token Lifecycle Management (TLCM) User Guide</i>                                              | TLCM is an application offered as part of VTS to assist clients to manage token lifecycle events and updates.                                                                                                       |
| <i>Visa Token Service HCE Wallet Provider Implementation Guide</i>                                    | Describes VTS capabilities and integration considerations for wallet providers.                                                                                                                                     |
| <i>Visa Token Service Implementation Guide for Issuer Participation in VDEP Third-Party Solutions</i> | Describes VTS capabilities and integration considerations for issuers participating with third-party wallets.                                                                                                       |



## 1 Visa Token Service Overview

The payments industry is evolving to support payment technologies that provide increased protection from counterfeit, account misuse, and other forms of fraud. While EMV<sup>®1</sup> chip cards can provide substantial protection for card-present transactions, a similar need exists to minimize unauthorized use of cardholder account data and to reduce cross-channel fraud in emerging transaction environments that combine mobile devices, e-commerce, and point-of-sale environments.

Tokenization holds substantial promise to address these needs. Payment tokens are surrogate values that replace Primary Account Numbers (PANs) stored electronically throughout the payments ecosystem and can be used to securely conduct payment transactions. To provide improved protection against misuse, tokens can be restricted for use in a specific domain, such as a device or channel. These underlying usage controls are a key benefit of payment tokens.

There are benefits for all stakeholders in the payments ecosystem that may help encourage adoption of payment tokens:

- Card issuers and cardholders may benefit from new and more secure ways to pay, improved transaction approval levels, and reduced risk of subsequent fraud in the event of a data breach in which payment tokens are exposed instead of the underlying PANs.
- Acquirers and merchants may experience a reduced threat of online attacks and data breaches, as payment token databases may be less appealing targets given their limitation to a specific domain. Acquirers and merchants may also benefit from the higher assurance levels that payment tokens offer.
- Payment processing networks have adopted a common EMV tokenization specification that facilitates interoperability and simplifies token processing for issuers, processors, and acquirers.

### 1.1 Payment Token Standard

A global payment token standard is intended to enhance the security of digital payments and simplify the purchasing experience when shopping on a mobile phone, tablet, personal computer, or other smart device. Issuers, merchants, and digital wallet providers can request a payment token so that when a cardholder initiates an online or mobile transaction, the payment token—not the traditional

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<sup>1</sup> EMV is a registered trademark or trademark of EMVCo LLC in the United States and other countries.

card account number that appears on the cardholder's plastic card—is used to authorize, clear, and settle the transaction in the same way traditional card payments are processed today.

The *EMV® Payment Tokenisation Specification Technical Framework* describes the payment token system landscape, the types of entities that provide key support for the use of payment tokens, the details to implement multiple use cases, and the benefits of adopting a unified approach. This framework supports acquirer, merchant, and issuer token implementations to provide enhanced data security for the cardholder PAN in payment transactions.

## 2 Visa Token Service Description

Visa Token Service allows issuers to support Visa payWave for mobile transactions. Historically, Visa payWave for mobile transactions required account credentials to be stored in a secure chip on the mobile device called the secure element. VTS allows issuers to provide their cardholders with mobile payment capabilities to make Visa payments with a range of Visa-approved HCE-capable mobile devices and HCE-based payment applications (e.g., Android and iOS) in addition to secure-element-capable mobile devices.

In the case of tokens issued for HCE-based NFC payments, issuers also play the role of the token requestor. In addition to the traditional activities around credential issuance and transaction processing, issuers supporting HCE payment programs can also develop mobile payment applications or add payment capabilities to existing mobile banking applications. This option requires them to connect to VTS as a token requestor in addition to integrating as a token issuer.

VTS provides the following functionalities to support Visa cloud-based payments:

- **Provisioning:** Generates a Visa token from the Visa Token Vault and provisions (delivers) the token to the token requestor.
- **Lifecycle Management:** Manages the token both at the Visa Token Vault and at the Mobile Application.
- **Active Key Management:** Based on the Visa cloud-based payments standards, VTS manages Limited Use Keys (LUKs) for Visa cloud-based payments use cases (e.g., HCE).
- **Token Transaction Processing:** Using the token-to-PAN mapping stored in the Visa Token Vault, VTS and VisaNet conduct token-to-PAN mapping, cryptogram validation, domain restriction check, and velocity checking (cloud-based payments tokens).
- **Post-Payment:** VTS matches transaction logs collected from a device against the transaction logs maintained in VisaNet to confirm that transactions are not made using a cloned instance of a payment application on another device. Post-payment verification is triggered on each LUK replenishment call and on demand by an issuer or Visa fraud monitoring system.

### 2.1 Service Capabilities

Tokenization for Visa cloud-based payments introduces four key interactions between token requestor (issuer), Visa, and issuer host:

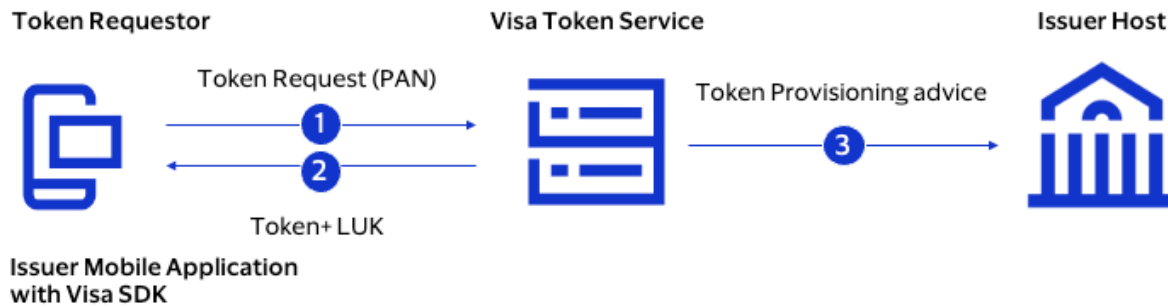
- Provisioning
- Transaction processing (authorization, full financial, and clearing messages)
- Lifecycle management

- Active key management

### 2.1.1 Provisioning

Figure 2–1 shows the token provisioning flow between the token requestor (issuer’s Mobile Application or Mobile Application Platform interacting with Visa’s SDK), Visa, and issuer host.

Figure 2–1: Provisioning Flow

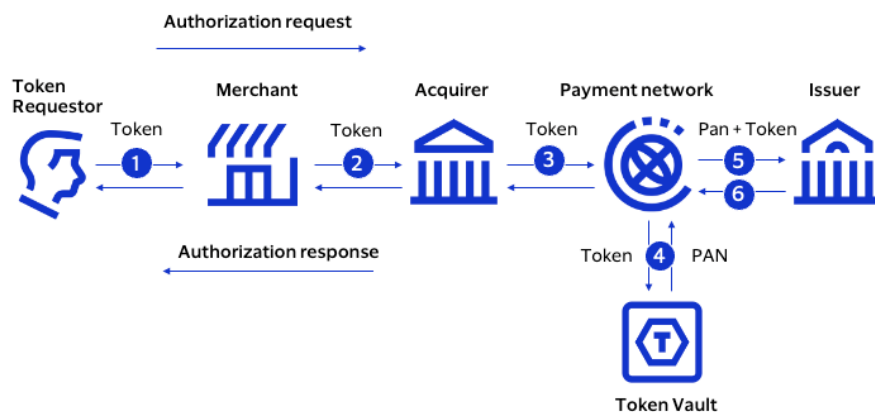


1. Token requestor submits a token request to VTS; VTS generates a token based on issuer-defined account management risk parameters to provision a token inside the issuer’s Mobile Application.
2. VTS generates a new payment token, creates a LUK, and provisions the token.
3. Visa can optionally send a notification to the issuer host to advise whether a token is provisioned.

### 2.1.2 Transaction Processing

Figure 2–2 shows the transaction processing flow.

Figure 2–2: Transaction Processing Flow



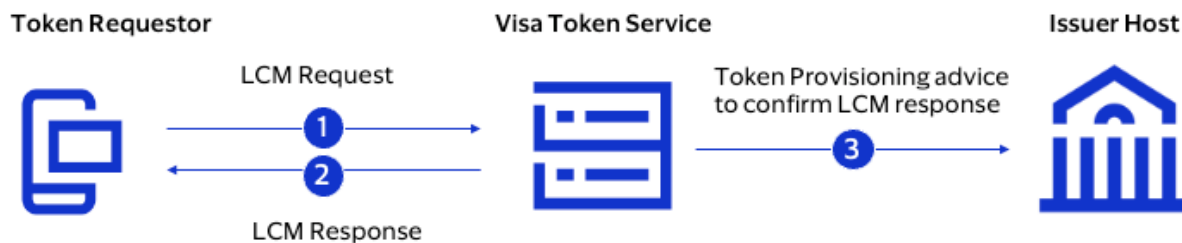
1. Consumer initiates a purchase at an NFC-enabled merchant.
2. Merchant submits a token in place of the PAN to its acquirer.
3. Acquirer passes the token (processed identically to the PAN) to VisaNet.

4. VisaNet detects the token and exchanges it with the PAN via VTS and performs LUK velocity checks.
5. VisaNet passes the PAN and token to the issuer for an authorization decision.
6. Issuer or its processor authorizes or declines the transaction and returns a response to VisaNet. VisaNet exchanges the PAN back to a token and sends a response to the acquirer and on to the merchant.

### 2.1.3 Lifecycle Management

Figure 2–3 shows the token lifecycle event flow between the token requestor, Visa, and issuer host.

Figure 2–3: Lifecycle Management (LCM) Flow



1. Token requestor requests lifecycle updates such as “suspend” or “delete/deactivate” of the payment token (for lost, stolen, or damaged devices).
2. Visa updates the Visa Token Vault with the status of the PAN-to-token mapping and notifies the token requestor.
3. Visa can optionally send a notification to the issuer host (0620 notification advice) with the status of the PAN-to-token mapping.

**Note:** The issuer or its processor can also initiate lifecycle requests for both token lifecycle and PAN updates (e.g., replacement, expiry date changes) through the Visa TLM tool on Visa Access.

### 2.1.4 Active Key Management

VTS enables issuers to configure several risk parameters, akin to velocity controls, such as number of transactions or time-to-live, that manage the triggers for refreshing LUKs to a provisioned device. To help ensure a consistent consumer experience in terms of performance and usability, VTS must implement a minimum set of parameters, configurable by service enrollment. The Active Key Management capability generates the initial set of LUK thresholds to deploy to the mobile device.

During transaction processing, if the service profile parameters for a particular account indicate that the LUKs on the device must be replaced, the Active Key Management capability recognizes this and alerts the token requestor that it should replenish account parameters.

Alternatively, if the on-device service profile parameters indicate that LUK replenishment is needed or is close to being needed (threshold management from the Mobile Application), then the Mobile Application can initiate account parameter replenishment.

## 2.2 Assumptions and Dependencies

- Transactions initiated with tokens generated and provisioned using VTS must be processed over VisaNet. In the U.S., VTS supports Visa debit transactions routed over certain unaffiliated PIN debit networks, such as Maestro and STAR.

**Important:** Within Europe, Visa Token Service can support any Visa-approved network processor providing authorization, clearing, and settlement. This document uses Visa processing for illustration purposes. For questions about Visa Token Service in Europe please contact your European relationship manager.

- Issuer host impacts need to be considered for each of the token payment use cases to be supported per the VisaNet Business Enhancements Global Technical Letter and Implementation Guide.
- Issuers acting as token requestors for their Android and iOS NFC mobile payment applications using HCE must perform their own identification and verification (ID&V).
- Issuers must integrate the a Visa certified SDK into their NFC Mobile Application for HCE payment programs.

## 2.3 Major Components

This section describes the major components of VTS for cloud-based payments and the role each plays for the generation, provisioning, processing, and lifecycle management of tokens.

Figure 2–4 shows how the major components interact for VTS.

The diagram illustrates the Visa Token Requestor API architecture. At the top, it states "Visa Token Requestor API\*" and a note: "(Issuer may use TR-TSP to connect to Visa)".

**Issuer Host:** Represented by a bank icon, it connects to the **VisaToken Service** via **ISO 8583**.

**VisaToken Service:** A central blue box containing:
 

- VisaToken Vault**
- Provisioning**
- Active Key Management**
- Lifecycle management**

 It connects to **VisaNet** via **ISO 8583**.

**VisaNet:** Represented by a Visa logo icon, it connects to the **Acquirer** (bank icon).

**Acquirer:** Connects to the **Merchant** (store icon).

**Merchant:** Connects to the **POS** (point-of-sale terminal icon).

**Issuer Wallet App:** A central box containing:
 

- Token Requestor Application Code** (e.g., Issuer's application code)
- Visa certified NFC Payment SDK**

 It connects to the **Token Requestor Mobile Application Platform** via a bidirectional arrow.

**Token Requestor Mobile Application Platform:** (e.g., Issuer's Mobile Application Server) connects to the **VisaToken Requestor APIs** via **REST**.

**VisaToken Requestor APIs:** A vertical blue bar that connects to the **Token Requestor Mobile Application Platform** via **REST** and to the **Issuer Host** via **ISO 8583**.

**POS:** Connects to the **Token Requestor Mobile Application Platform** via a bidirectional arrow.

- Visa Token Service
- Visa certified NFC Payment SDK
- Issuer's Mobile Application Platform
- Issuer's Mobile Application
- Issuer host

### 2.3.1 Visa Token Service

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provision the account as a Visa cloud-based payments account, so a token and a LUK are provisioned (VTS subsequently updates the LUK via the Active Key Management capability).

Because tokens for NFC HCE applications are provisioned as Visa cloud-based payments accounts, issuers must establish a new subscription profile during enrollment. VTS enables issuers to configure a subscription to allow them to input the desired risk parameters for the accounts in that subscription. These parameters include velocity controls, such as number of transactions or time-to-live, that manage the triggers for refreshing the LUK on a provisioned device.

Further, issuers developing and managing their own NFC HCE payment application must integrate to VTS as token requestors. The issuer's Mobile Application Platform accesses VTS through application programming interfaces (APIs) designed to process provisioning requests, lifecycle management, and Active Key Management events.

VTS also stores token-to-PAN mappings and manages all domain restrictions associated with each token provisioned.

When a transaction is conducted with a token, the following actions occur:

- Token cryptogram is validated
- Appropriate velocity checks are conducted (HCE use case)
- Domain restrictions are checked
- Token is mapped to the PAN
- Results are forwarded to the issuer host

Issuers can access data in the VTS using new web service interfaces and traditional ISO8583 messages. Issuers can also execute token lifecycle management events through the Visa TLM tool on Visa Access.

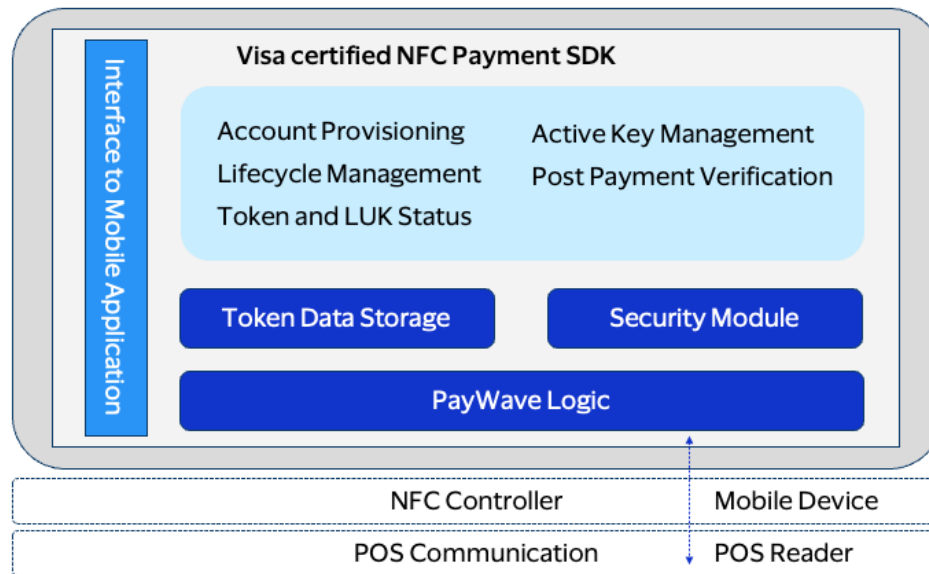
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### 2.3.2 Visa certified NFC Payment SDK

The Visa certified NFC Payment SDK provides a framework on which issuers can create payment experiences using VTS. Specifically, issuers can use a Visa provided, a Visa partner provided or develop and certify their own NFC Payment SDK for HCE-based payments at the point of sale (POS).

Figure 2–5 shows the relationship between the major components and the entities that provide tokens and process payments.

Figure 2–5: Visa certified NFC Payment SDK



The SDK offers the following capabilities:

- Interface for the Mobile Application to communicate with SDK components
- Device-to-POS terminal communication using Visa payWave standards
- Storage management to manage tokens and their account parameters
- Security of sensitive token data

The Visa provided NFC Payment SDK is built for the Android and iOS platform to help issuers develop their own Android or iOS payment applications using HCE. A Visa certified NFC Payment SDK resides in the Mobile Application on the Android or iOS device.

### 2.3.3 Issuer's Mobile Application Platform

The Mobile Application Platform is the server that the issuer uses to communicate to the Mobile Application on an Android device. Typically, the Mobile Application Platform provides functions such as:

- Authenticate the accountholder (e.g., online banking credentials)
- Communicate account balances
- Send card art
- Send the Mobile Application alerts or notifications

- Interface with the Visa Digital Solutions APIs. Orchestrate the data exchange between Mobile Application and VTS. Depending on the solution architecture the MAP may connect to VTS indirectly via the services of a Token Requestor TSP.

The following describes the primary functionality issuers (or their TR TSP service provider) need to develop on their Mobile Application Platform (MAP):

- SDK initialization
  - Mobile Application requests device key pair from MAP. Receive device identity from the Mobile Application.
  - MAP sends encrypted Device Symmetric Key (DSK) to the Mobile Application. (DSK is a symmetric key, encrypted by the MAP using an Issuer Bootstrap Key (IBK) public key).
  - MAP sends device certificate and device RSA key pairs (encryption and signature keys) to the Mobile Application.
  - MAP generates a device ID using device information and passes it to the Mobile Application.
- Enroll device
  - MAP generates the keys and certificates for encrypting and signing data in transit from VTS to the device.
  - MAP gets device attributes to send in the Enroll Device call from the Mobile Application. The Mobile Application gets the device attributes from the SDK.
  - MAP enrolls the device in VTS.
- Enroll PAN
  - MAP sends a list of eligible cards to the Mobile Application. The Mobile Application displays a list of eligible cards to be provisioned. The cardholder selects a card, and the card information comes back to the MAP.
  - MAP calls Enroll PAN API to enroll the PAN in VTS.
  - VTS responds back with vPanEnrollmentID. The MAP must store the mapping between the PAN and vPanEnrollmentID.
  - Issuers can explicitly opt to not receive 0100 TAR or 0100 AV messages for push provision requests initiated by their own issuer wallet by setting an indicator in the encrypted payment instrument to not require ID&V.
- Token enrollment
  - MAP calls the Provision API by passing vPanEnrollmentID, device identifier, cardholder, and optional risk information.
  - VTS responds with the vProvisionedTokenID, encrypted token, encrypted LUK, and other personalization data.
  - MAP passes the token and LUK data to the Mobile Application.
- Lifecycle management

- MAP submits a lifecycle event to VTS to change the status of a token.
- VTS modifies the token status and notifies the token requestor and token issuer that the token status has changed.
- Active Key Management
  - Mobile Application gathers data from the Visa Digital Solutions SDK to replenish the LUK on the device.
  - MAP receives the data from the MAP call and sends a replenishment request to VTS.
  - VTS sends a new LUK and related parameters to the MAP. The MAP forwards the VTS response to the Mobile Application.
  - Mobile Application stores the updated LUK in the SDK and sends the acknowledgement to the MAP. The MAP forwards the acknowledgement to VTS.
- Notification
  - VTS sends the notifications of different events to the MAP server.
  - The notification URL is configured in the Visa Developer Center.
  - Active Key Management (LUK replenishment) notifications, token creation notifications, and token activation notifications are sent from VTS to the MAP.

---

### 2.3.4 Issuer's Mobile Application

The issuer's Mobile Application provides the user interface experience for the cardholder on Android or iOS devices that support HCE-based NFC payments. The cardholder uses the Mobile Application to enroll accounts into the service, initiate payments, and manage those accounts over their lifecycle.

In addition to the user experience, the issuer's Mobile Application also provides logic to facilitate communications between the issuer's MAP and a Visa certified NFC Payment SDK. Interactions between the issuer's Mobile Application and the issuer's MAP are proprietary to the issuer's implementation.

The following describes the primary functionality that issuers need to develop on their Mobile Application:

- User interface and user experience to interact with payment instruments
- Methods to verify cardholder
- Management of card art and other metadata
- Implementation of the Visa Developers SDK interface, which includes SDK initialization, store and manage tokens, and store and manage LUK. For more information on SDK interfaces, issuers should refer to the documentation of their respective SDK provider.

### 2.3.5 Issuer Host

The issuer host plays its usual role of being responsible for risk management and payment authorizations. The issuer host also has the option to submit lifecycle events and receive notifications regarding token provisioning.

## 2.4 Tools and Resources

To enable the tokenization of its payment accounts, the issuer needs to integrate into VTS as a token issuer. This consists of defining product eligibility, setting up token BINs, and establishing token service configurations. Issuers developing and managing their own NFC HCE payment application for Android or iOS must also integrate to VTS as a token requestor. The issuer's MAP accesses VTS through APIs designed to process provisioning requests, lifecycle management, Active Key Management, and other events.

Service configuration and program management as a token issuer is managed through a set of VTS enrollment tools, which are available through Visa Access. For more information, please see Appendix A, Token Lifecycle Management.

Service configuration and program management as a token requestor is managed through the Visa Digital Solutions Developer Center (Sandbox) and Visa Digital Solutions Business Center (Production). For more information regarding using the Visa Digital Solutions Business Center, please see Appendix B, Visa Digital Solutions Developer Center.

**Table 2-1: VTS Onboarding Resources**

| Tool                       | Activity                                                                                                                                                                                                                                                              |
|----------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visa TLCM Tool             | <ul style="list-style-type: none"> <li>Accessed through Visa Access: <ul style="list-style-type: none"> <li>Token inquiry</li> <li>Delete/deactivate token</li> <li>Suspend token</li> <li>Resume token</li> <li>Update PAN or PAN expiry date</li> </ul> </li> </ul> |
| Visa Resolve Online (VROL) | <ul style="list-style-type: none"> <li>Optional: Accessed through Visa Access: <ul style="list-style-type: none"> <li>Token transaction exception processing for retain and return obligation (if applicable).</li> </ul> </li> </ul>                                 |

## Visa Token Service Implementation Guide for Participant-Branded Solutions (Host Card Emulation—Issuer Wallet): Visa Supplemental Requirements

| Tool                                              | Activity                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visa Digital Solutions Developer Center (Sandbox) | <ul style="list-style-type: none"><li>• Visa configures on the developer portal:<ul style="list-style-type: none"><li>– Create issuer admin developer account</li><li>– Create initial issuer Mobile Application profile</li><li>– Generate Visa Digital Solutions API key and shared secret</li></ul></li><li>• Issuer configures on developer tool:<ul style="list-style-type: none"><li>– Add additional issuer developer accounts</li><li>– Create additional Mobile Application profiles</li><li>– Generate additional Visa Digital Solutions API keys and shared secrets</li></ul></li></ul> |
| VCMM                                              | The VCMM tool is an online tool accessible via the Visa Access portal to help manage card art configuration, terms and conditions, and other metadata at account range level for tokenized BINs. The configured card art images, terms and conditions, and metadata are used by the digital wallet provider to render during token provisioning.                                                                                                                                                                                                                                                   |



## 3 Integration Process

This chapter describes the process for integrating with VTS.

### 3.1 Agreements

Table 3–1 describes the Visa contracts that are required for issuers to implement VTS. Issuers should contact their Visa representative for details about these agreements.

Table 3–1: Agreements

| Agreement                                         | Description                                                                                                                                              |
|---------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visa Token Service Issuer Participation Agreement | Executed between the issuer and Visa, this agreement describes the obligations, rights, and responsibilities of Visa and the issuer with regards to VTS. |

### 3.2 Forms (Sandbox)

Table 3–2 describes the forms needed for issuers to integrate with VTS in the sandbox environment. Issuers should contact their Visa representative to obtain these forms.

Table 3–2: Forms Needed for Integrating with VTS in Sandbox Environment

| Form                                                     | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
|----------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Visa Token Service Issuer Program Information Form (PIF) | This form is intended for issuers to provide Visa with information relating to their VTS program launch plan.                                                                                                                                                                                                                                                                                                                                                                                             |
| Client Information Questionnaire (CIQ)                   | This form is submitted to the Client Configuration Management analyst and used for token BIN(s) and Processor Control Record (PCR) setup, and Key Conveyance. The issuer has the primary responsibility of filling out this form. The Visa Customer Support Service team may help the issuer fill out this form as needed. This form may not be required per the current Visa payWave issuance status of the issuer. For further details on this topic, issuers should contact their Visa representative. |
| Developer Center Onboarding Form                         | The issuer completes the Developer Center Onboarding Form to get access to the API keys and shared secrets and other Mobile Application profile information.                                                                                                                                                                                                                                                                                                                                              |

| Form                                | Description                                                                                                                                                                                                                            |
|-------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Global Client Testing Questionnaire | The issuer completes this questionnaire to submit a service request to Global Client Testing (GCT) for the Client Card Personalization Review and to schedule host testing. This form applies to all Visa payWave for mobile projects. |
| VTS Token Requestor Onboarding Form | This form provides a list of elements required for issuers to integrate with VTS as token requestor. Visa assigns a token requester ID to the issuer.                                                                                  |

### 3.3 Forms (Production)

Table 3–3 describes the forms needed for issuers to implement VTS. Issuers should contact their Visa representative to obtain these forms.

**Table 3–3: Forms Needed for Integrating with VTS in Production Environment**

| Form                             | Description                                                                                                  |
|----------------------------------|--------------------------------------------------------------------------------------------------------------|
| Developer Center Onboarding Form | The issuer completes this form to get access to the API keys, shared secrets, and other profile information. |

### 3.4 Pre-Implementation Tasks

This section describes the tasks that must be completed before implementation can start.

#### 3.4.1 Identify the Implementation Team

The issuer assembles the team of its personnel necessary for the implementation project. After the issuer and Visa project teams have been formed, a formal kickoff meeting is scheduled between the key members of these two teams to understand the roles and responsibilities of each team member.

#### 3.4.2 Project Scoping

The core issuer and Visa project teams collaborate to perform gap analysis between the current capabilities and the new capabilities that may be added to the issuer's system. As a prerequisite, technical members of both project teams should review the documents described in Table 1, Related Publications, prior to the scoping session. Scoping sessions should include an in-depth analysis of issuer requirements, including but not limited to:

- High-level business scope

- Assumptions, constraints, and dependencies
- High-level business context
- Model and description of payment use cases
- Payment credentials provisioning and management use cases
- Overall system architecture
- Reporting requirements
- Non-functional requirements
- Risk and open issues

The Visa project team uses the VTS Issuer PIF described in Section 3.2 to get a detailed understanding of the issuer's project objectives. Both Visa and the issuer are responsible for making the necessary resources available to complete this phase of the integration process.

Please note that in case an Issuer TSP and/or Token Requestor TSP are involved, some of the activities mentioned in these paragraphs may be performed by the Issuer TSP an/or Token Requestor TSP in conjunction with, or on behalf of the Issuer.

---

#### 3.4.2.1 Use Case Analysis

The Visa and issuer project teams should perform an in-depth analysis of the use cases. VTS provides support for the standard set of use cases that should meet the requirement of any issuer.

Using VTS, an issuer can support use cases, including but not limited to the following:

- Add Token: Add a token to a Mobile Application.
  - Adding a token and processing the 0100 TAR/AV message
  - Optionally adding a token without receiving the 0100 TAR/AV message
- Tap & Pay Transaction: Make an NFC transaction using the token provisioned in the Mobile Application.
- Delete/Deactivate Token: Delete or deactivate a token on a Mobile Application.
- Update PAN: Update token-to-PAN mapping or token configurations in scenarios, including but not limited to:
  - Primary plastic card has expired and re-issued card needs to be mapped to existing token.
  - Cardholder's provisioned card is lost or stolen and a new card needs to be linked to the token on the device.
  - Card's token or PAN expiration date needs to be updated.
- Suspend Token: Suspend the token if for any reason the issuer wants to temporarily block a cardholder's provisioned mobile payment token.

- **Resume Token:** Resume the token if for any reason the issuer wants to unblock a cardholder's previously suspended mobile payment token.
- **Active Key Management:** The periodic replenishment of LUKs required for NFC payment tokens using HCE.
- **Card Metadata Update:** If the issuer uses VTS to manage card metadata this use case defines the flow of updating card metadata after a card is provisioned on a device.
- **Token Transactions Notification:** Displays transactions cardholders make using token credentials on the issuer's Mobile Application.

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#### 3.4.2.2 Issuer Host Considerations

Token transaction processing describes all aspects of processing, including how token-related data is passed and processed in authorizations, clearing, settlement, and dispute management. Issuers that participate in VTS have to make changes to their existing system to process token-related transactions.

VTS requires the use of token-related data elements in existing fields and the introduction of new fields and values to process token transactions and help ensure interoperability within the payments ecosystem. Issuers must be prepared to receive both the payment token and cardholder PAN in authorization and full financial messages and clearing transactions and must include both the payment token and PAN in chargebacks, chargeback reversals, and requests for copy.

Verification methods for VTS for NFC payments using HCE are facilitated using the following profiles and acceptance methods:

- **MSD Verification Value** for magnetic stripe data (MSD)  
**Note:** MSD Verification Value may not be applicable to all regions
- **CVN 43** for quick Visa Smart Debit/Credit (qVSDC)

When a token is used to initiate a transaction for NFC payments using HCE, chip data (including an application cryptogram) is included in the transaction message. Issuers have the option of selecting one of following message types:

- **Full Chip:** A Visa Smart Debit/Credit (VSDC) implementation option in which the issuer has made all host system changes needed to transmit and receive chip data (e.g., Field 55) in V.I.P. Clearing and Settlement messages.
- **Token Convert:** Eliminates chip data-related host system changes by removing chip data prior to sending the authorization request to the issuer and Field 44.5 contains a result of the MSD Verification Value. For those issuers that do not support chip data, this option validates and converts the token transaction cryptogram to appear as a PAN-based, magnetic stripe transaction for HCE Visa payWave NFC transactions.

**Note:** Visa recommends that non-chip issuers select the Token Convert processing option.

In order to minimize the changes required in the issuer host system to support token transactions, Visa provides token processing support to issuers. Visa facilitates token payment transaction processing for all issuers irrespective of their chip processing capabilities. Visa helps ensure issuers receive relevant full chip data or Token Convert data, depending on their processing capabilities. The main objective of token processing is to validate token transaction data and to insert and send appropriate PAN transaction data to the issuer in the authorization request. As part of token processing Visa always:

- Performs validation on behalf of the issuer using MSD Verification Value or CVN 43. If the validation fails, Visa declines the request message and forwards a stand-in processing (STIP) advice to the issuer.
- Applies token domain restrictions and velocity checks.

Issuers that support HCE mobile devices receive specific data (ISO fields) in POS transactions from qVSDC terminals or MSD terminals depending upon the processing option selected by the issuer. Issuers should refer to Table 1, Related Publications, for the issuer-related changes needed to make to payment token transactions.

#### 3.4.2.2.1 Authorization

---

Visa sends fields and values in POS authorization and full financial request messages to the issuer, and the extent of the issuer impact varies by the types of transactions that are supported (mobile contactless Visa payWave transactions). Issuers need to support several token-specific fields and data in the authorization messages in order to participate in this service.

In order to minimize the changes required in the issuer host system to support chip-based NFC Visa payWave and e-commerce token transactions, Visa will implement Token Convert messages to convert all processed data based on the PAN. To do this Visa:

- Validates the cryptogram and removes the non-relevant chip data
- Generates and provides a CVV/ICVV/CAVV validation result
- Converts the chip-initiated token transaction to a magnetic stripe-based transaction and sends the authorization request to the issuer

Issuers must be aware that when participating in this service, Visa always performs validation on behalf of the issuer using MSDVV or CAM.

Depending on the issuer setup option or the acceptance method at the NFC POS, Visa forwards the verification results code in Field 44.5—CVV/ICVV Results Code.

#### 3.4.2.2.2 Clearing and Settlement

---

Clearing and settlement flow from the acquirer to VisaNet, and VisaNet to card issuer, is impacted by participation in VTS. BASE II sends token data and the cardholder PAN in POS clearing transactions to the issuer. Issuers receive the POS clearing transactions and process them accordingly.

The following changes need to be supported by issuers upon participation in VTS.

- Draft Data, TC x5 and TC x6, transaction code qualifier (TCQ) 0, TCR 5 Payment Service Data, positions 150–165, PAN token, which will include token data
- Redefined BASE II fields to support Draft Data, TC x5, TCQ 0, and TC x6, TCQ 0, TCR 1—Additional Data, positions 6–7, which will contain the token assurance level value
- TC 40—Fraud Advice Transactions, TCR 2, positions 146–161, PAN token
- TC 40—Fraud Advice Transactions, TCR 2, positions 162–164, PAN token Extension
- New BASE II value of 4 (Visa-generated token data) in the existing TC 48—BASE I Advise Record, TCR 6, position 82
- TC 52—Retrieval Request Record Transaction, TCR 1, positions 88–103, PAN token Redefined BASE II fields to support Draft Data TC x5 TCQ 0, and TC x6, TCQ 0, TCR 1—Additional Data, position 5; that will contain a space
- Redefined BASE II fields to support Draft Data, TC x5, TCQ 0, and TC x6, TCQ 0, TCR 1—Additional Data, positions 6–7, Token Assurance Level

#### 3.4.2.2.3 Exception Processing

---

The token is a “retain and return” value for exception item processing. Issuers must retain and return the cardholder PAN in the account number field and token values in the token fields, and submit them in chargebacks, chargeback reversals, and request for copies for tokenized transactions.

Issuers must retain and return both:

- The cardholder PAN from the TCR 0, positions 5–20
- PAN and the token from the TCR 5, positions 150–165, for exception processing in chargebacks and chargeback reversals

#### 3.4.2.3 Token Provisioning Decision

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Issuers need to determine the approach for verifying the identity of the cardholder before a token is provisioned. For NFC HCE payments in issuer Mobile Applications, the issuer is responsible for performing cardholder ID&V. VTS provisions active tokens upon direct request from an authenticated issuer's Mobile Application.

Under normal scenarios the 0100 TAR/AV message will be sent to the issuer host for processing during token provision.

Issuers can optionally suppress the 0100 TAR/AV message by explicitly indicating this during the token provisioning request call. This can be achieved by setting the PAN source field to issuer push provision and encrypted payment instrument ID&V optional flag to false.

---

#### 3.4.2.4 Active Key Management Considerations

Issuers implementing Android or iOS NFC mobile payment applications using HCE must configure parameters that define the validity of the LUKs during token transaction processing and the frequency with which LUKs are replenished on the consumer device.

Issuers need to make choices for the following LUK configuration options:

- **VisaNet-Level Thresholds:** Define the risk profile of token processing. VisaNet tracks the time-to-live, number of transactions, and cumulative spend for both the current and previously provisioned LUKs. Issuers are able to differentiate between thresholds for domestic and international transactions and between transactions when the issuer host system is available or when Visa is doing stand-in processing.
- **Device-Level Thresholds:** Define usage parameters that need to be tracked by the Mobile Application. These Mobile Application-level usage parameters define how often the Mobile Application connects to VTS to request replenishment of the LUK. The Mobile Application must track time-to-live and number of transactions per LUK.

Visa recommends that the LUK replenish should be triggered by the Mobile Application after reaching the device-level thresholds. Device thresholds should be lower than the VisaNet thresholds.

**Note:** Please refer to Section 3.5.1.8, Define Issuer LUK Replenishment Parameters, for examples.

---

#### 3.4.2.5 Lifecycle Management Considerations

Issuers or token requestors need to determine how to support token lifecycle events. Token lifecycle interactions (e.g., suspend/resume, update) can be initiated by the:

- Issuer through the TLCM tool on Visa Access.
- Issuer through ISO messages.
  - Issuers can send ISO 8583 0302 token maintenance messages to manage the status of tokens.
- Token requestor through lifecycle APIs exposed by VTS.

### 3.4.2.6 Issuer Alerts and Notifications

Issuers have the option to receive the following notifications for token provisioning, lifecycle management, authorized transactions, and Active Key Management events.

- Issuer notifications for token provisioning:
  - Issuers have the option to receive token advice messages (ISO 8583 0620 messages) following a token provisioning event. Issuers can subscribe to these messages during the VTS enrollment process.
- Token requestor notifications for lifecycle management:
  - After lifecycle events, token requestors (including issuers acting as token requestors for their NFC mobile payment applications using HCE) have the option to receive lifecycle management notifications through web service notifications. Token requestors need to subscribe to these messages during token requestor onboarding.
- Token requestor notifications for Active Key Management:
  - When integrated to VTS through a MAP, issuers acting as token requestors for their Android or iOS NFC mobile payment applications using HCE have to provide an interface to receive LUK “threshold exceeding” notifications from VTS. After receiving this notification, token requestors must initiate LUK replenishment through VTS.
- Token requestor notifications for token transactions:
  - When integrated to VTS through a MAP, issuers acting as token requestors for their Android or iOS NFC mobile payment applications using HCE have to provide an interface to receive transaction notification. After receiving this notification, token requestors may call VTS to get details on the cardholder’s transaction. Token requestors need to subscribe to these messages during token requestor onboarding.
- Token requestor notifications for card metadata update:
  - When integrated to VTS through a MAP, issuers acting as token requestors for their Android iOS NFC mobile payment applications using HCE have to provide an interface to receive card metadata update notification. The issuer can choose to store and manage card metadata through VTS, which sends the initial and/or updated card metadata to the issuer’s Mobile Application. If the issuer manages the card metadata on its MAP, then these notifications are not triggered from VTS.

### 3.4.2.7 Environments for Integrating as a Token Requestor (Mobile Application)

Visa, as part of a VTS implementation, provides the following environments that the issuer should prepare to integrate with:

- Visa Digital Solutions Developer Center (Sandbox) – This is the validation/quality assurance (QA) environment, which allows token requestors (including issuers acting as a token requestor via their

Mobile Application or MAP) to perform end-to-end testing with VTS before they roll out their application in the production environment.

- Visa Digital Solutions Business Center (Production) – After the token requestor's integration with VTS is thoroughly tested in the validation environment, the solution is deployed in the production environment.

---

#### 3.4.2.8 Visa Provided SDK Considerations

Visa, as part of a VTS implementation, can offer a Visa provided SDK to Mobile Application developers but does not mandate the usage of this SDK. If issuers choose to use a third-party SDK or develop it in house, then the issuer is required to obtain the Visa payWave and Mobile Application security compliance through a third-party lab or tool.

---

#### 3.4.2.9 Mobile Application Platform (MAP) Development Considerations

Mobile Application (MA) developers are required to develop a MAP to orchestrate communication between their MA and VTS. The design and architecture of this component is the issuer's responsibility. The definition and implementation of the interface between the MAP and MA is also the issuer's responsibility.

Visa, as part of a VTS implementation, can offer a Visa Provided Token Requestor TSP service to Mobile Application developers but does not mandate the usage of this service. If issuer chooses to use a third party Token Requestor TSP the above mentioned responsibilities and any certification requirements would be the Token Requestor TSP responsibility. The definition and implementation of the interface between the MAP and Token Requestor TSP is the responsibility of the Token Requestor TSP service provider.

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#### 3.4.2.10 Access Security Considerations

VTS uses a layered security approach to protect access to VTS services and to help ensure that communication is secure. Layered security, also known as layered defense, is the practice of combining multiple mitigating security controls to protect resources and data.

The VTS layered approach is composed of transport-level security. The issuer (or Token Requestor TSP and Issuer TSP on the issuers behalf) can select to use either one-way Secure Socket Layer (SSL) or mutual SSL to assure that the endpoint can trust the communication with VTS and that the transport channel is encrypted.

### 3.4.3 Project Planning

The output of scoping sessions should be taken as input to build a detailed scope of the project. This scope is then reviewed and validated by the issuer and Visa VTS project team members. The Visa Token Service Issuer PIF can be used to capture scope for a VTS project. The VTS implementation team is responsible for preparing the PIF. The PIF represents common and agreed-upon understanding between Visa and the issuer of the scope of the VTS project.

The VTS project manager builds the project plan and shares it with the issuer. This is followed by:

- A meeting between the VTS project manager and the issuer project team so the VTS project manager can provide an in-depth review of the VTS project plan.
- A meeting between the VTS project manager and the issuer project manager to synchronize the VTS project plan with the issuer's project plan. This includes aligning the dates of key deliverables from Visa that are dependent on action from the issuer and vice versa.
- The issuer signs off on the project plan.
- The issuer's commitment to complete the tasks assigned to it on time and in their entirety.

## 3.5 Sandbox Implementation Tasks

This section describes the sandbox implementation tasks for token requestors and the issuer host.

### 3.5.1 Implementation Tasks as Token Issuer

This section describes the following sandbox onboarding and integration tasks for token requestors:

- Complete required issuer host changes
- Define product eligibility
- Set up token BINs
- Set up card metadata
- Issuer host key management
- Define lifecycle management
- Set up notifications and alerts
- Define issuer LUK replenishment parameters

### 3.5.1.1 Complete Required Issuer Host Changes

Issuers need to make changes to their host systems as a prerequisite for participation and integration to VTS. Issuers should follow business-as-usual procedures for making and testing changes. For more information, issuers should contact their Visa representative.

### 3.5.1.2 Define Product Eligibility

Visa requires PAN BIN and PAN range information from the issuer to perform a product eligibility check for provisioning. The issuer needs to provide the following information about PAN BIN and PAN range.

- Issuer Business ID (BID)
- Product type(s) (e.g., credit, debit, prepaid)
- PAN BIN
  - BIN range start
  - BIN range end

### 3.5.1.3 Set Up Token BINs

VTS generates tokens from dedicated token BIN ranges that are mapped to PAN BIN ranges the issuer provides. Table 3–4 provides a sample of how a PAN BIN range maps to a token BIN range for token generation. A Visa representative performs this step in the sandbox environment. The issuer is required to provide the list of BINs to be configured using the CIQ.

Table 3–4: PAN BIN Range to Token BIN Range Mapping Example

| PAN BIN Range (Example) | Token BIN Range (Example) |
|-------------------------|---------------------------|
| 411111 001              | 444444 001                |
| 411111 901              | 444444 002                |
| 422222 001              | 444444 003                |
| 422222 100              | 444444 004                |
| 433333 00...            | 444444 005                |

### 3.5.1.4 Set Up Card Metadata

The issuer is responsible for generating card metadata and can either manage the metadata on the application or use the VCMM tool. Visa configures the data on behalf of the issuer in the sandbox environment. Typical metadata includes card art images, terms and conditions, and back of the card

details. For more details about VCMM, please refer to Table 2–1, VTS Onboarding Resources. In the sandbox, default card metadata is used.

---

### 3.5.1.5 Issuer Host Key Management

Visa requires a Master Derivation Key (MDK) to create and provision tokens and to validate token transaction cryptograms during token processing. Issuers have the following options for generating MDKs for VTS:

- Visa creates new MDKs and holds them (not transmitted to the issuer). This option is recommended for issuers that do not have existing chip programs.
- Visa can use existing issuer MDKs already on file with Visa. This option is recommended for existing chip issuers.
- Issuer creates new MDKs and transmits them to Visa via an existing Zone Control Master Key (ZCMK).
- Visa creates new keys and transmits them to the client via an existing ZCMK.

Issuers use VTS to establish key management procedures. Issuers should refer to Appendix A, Token Lifecycle Management, for more details.

---

### 3.5.1.6 Define Lifecycle Management

Token requestors (issuer's MA, MAP) have the ability to perform lifecycle management on their tokens to help ensure that VTS reflects the current status of their tokens. The status of a token can change at any point during its issuance, for example, due to an update to the PAN expiry date or a fraud alert on the PAN. Maintenance of these lifecycle events is important and helps ensure the integrity of the payment transaction.

The functions of lifecycle management are:

- Delete/deactivate token
- Suspend token
- Resume token
- Token inquiry
- Update token expiration
- Replace PAN mapped to token

The following options for lifecycle management are available:

- Supporting lifecycle management through the TLCM tool: Issuers can subscribe to use the TLCM tool during the VTS enrollment process to manage lifecycle events (required). Appendix A, Token Lifecycle Management, shows the enrollment process for the TLCM tool.

- Supporting lifecycle management through ISO messages from the issuer host: If using ISO messages, issuers need to support the ISO 8583 0302 messages.
- Supporting lifecycle management through the Visa Account Updater Suite (VAUS), a suite of solutions that provide e-commerce and card-on-file (COF) credential updates across PANs and network tokens.
- Supporting lifecycle management through web services from the token requestor: The MAP needs to directly implement the lifecycle management APIs exposed by VTS. The issuer has to ensure it is collecting all the required data to generate the appropriate message payload required by the API. Issuers should refer to the Visa Digital Solutions Developers Guide for the appropriate APIs for TLM.

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#### 3.5.1.7 Set Up Notifications and Alerts

Issuers must subscribe to and/or develop support to receive desired notifications for token provisioning, lifecycle management, Active Key Management, and other events. Issuers need to select from the options listed in Section 3.4.2.6, Issuer Alerts and Notifications.

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#### 3.5.1.8 Define Issuer LUK Replenishment Parameters

Issuers implementing Android and iOS NFC mobile payment applications using HCE must configure parameters that define the frequency with which LUKs are replenished. A Visa representative performs the sandbox configuration on your behalf.

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##### 3.5.1.8.1 VisaNet-Level Processing Thresholds

The issuer needs to define the LUK risk parameters that are tracked by VisaNet. These parameters define the risk profile of an LUK. VisaNet tracks the LUK velocity parameters based on incoming authorization requests and compares against the issuer-defined thresholds. When the parameters reach the threshold, VisaNet either declines the transactions or sends them to the issuer with the appropriate flags in the 0100 auth message.

The issuer is required to configure the following:

- Number of transactions made
- Time-to-live
- Cumulative spend

VisaNet tracks the parameters against the current and one previous LUK provisioned on the device. It also tracks parameters separately for domestic and international transactions. Please refer to Figure 3–1, Visa LUK Thresholds Guidelines, for a complete list of parameters and recommendations.

**Number of transactions (NOT)** is the number of transactions that should be processed with an LUK before it must be replenished. (e.g., 30).

**Time-to-live (TTL)** is the time-to-live (in days) for an LUK before it must be replenished. (e.g., 28 days).

**Cumulative Transaction Amount (CTA)** is the maximum cumulative amount that can be spent or authorized with an LUK before the LUK must be replenished (e.g., \$500).

**Current LUK settings** are used to do velocity checking on transactions based on the latest LUK provisioned on the device.

**Previous LUK settings:** The previous LUK thresholds are used against the LUK that was provisioned prior to the current LUK on the device. Previous LUK thresholds are required to process transactions that reach VisaNet after the LUK was updated on the device. For example:

1. A consumer makes a transaction “A” at 9:00 a.m. using LUK1 provisioned on the device.
2. Right after the transaction, the device initiates the process to replenish LUK1. LUK2 is replenished at 9:01 a.m.
3. Transaction “A” reaches VisaNet at 9:04 a.m. The delay could be due to a network glitch at the merchant or acquirer, or because the merchant is not processing the transaction in real time.
4. At the time of processing, VisaNet is aware that the device has LUK2 provisioned, so instead of using the threshold for current settings, VisaNet uses the thresholds configured for the previous LUK to process the transaction.

VisaNet only keeps track of the current LUK and one LUK prior to current LUK. Any transaction based on an LUK prior to the previous one is declined at VisaNet.

**Issuer available:** If the issuer is available to process the transaction, then VisaNet uses issuer available LUK thresholds to process the transaction.

**STIP:** If the issuer is not available to process the transaction, then VisaNet uses STIP LUK thresholds to process the transaction.

**Domestic:** All domestic transactions use domestic LUK thresholds for velocity checking. Domestic and international transactions are counted separately (i.e., the NOT counter for domestic and international are separate counters).

**International:** All international transactions use international LUK thresholds for velocity checking. Domestic and international transactions are counted separately. For example, CTA for domestic and international transactions are counted separately.

VisaNet performs LUK velocity checks for token-based transactions against the configured threshold. If any of the thresholds are exceeded, then VisaNet sends a 0100 auth message to the issuer host with

tags 1f31, 1f32, 1f33 in Field 123 to indicate which LUK threshold is exceeded. The issuer host makes the final decision to approve or decline the transaction.

VisaNet also offers an option to decline a transaction on the issuer's behalf if any of the thresholds are exceeded, even if the issuer host is available to process. Issuers should contact their Visa representative for more details on this configuration option.

## 3.5.1.8.2 Device-Level Thresholds

The issuer also must define usage parameters that need to be tracked by the MA. These Mobile Application-level usage parameters define how often the MA connects to VTS to request replenishment of the LUK. The MA must track the following per LUK:

- Number of transactions made
- Time-to-live

## 3.5.1.8.3 Visa Threshold Recommendations

Visa recommends that the LUK replenish should be triggered by the MA after reaching the device-level thresholds. Device thresholds should be lower than VisaNet thresholds. Figure 3–1 outlines guidelines for VTS LUK parameters.

Figure 3–1: Visa LUK Thresholds Guidelines

|                                                                                                                                              | Luk Parameters   | Valid Values For Current                                                                                            | Valid Values For Previous | Default Values For Current | Default Values For Previous | Comments                                                                                                                                                   |
|----------------------------------------------------------------------------------------------------------------------------------------------|------------------|---------------------------------------------------------------------------------------------------------------------|---------------------------|----------------------------|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>VIP</b> <ul style="list-style-type: none"><li>• Must be <b>same</b> across wallets</li><li>• Can be set at BIN / Acct. Range</li></ul>    | TTL (# of days)  | >=18                                                                                                                | N/A                       | 30                         | N/A                         | Time to live in days                                                                                                                                       |
|                                                                                                                                              | NOT* (#)         | >=10                                                                                                                | >=1                       | 10                         | 1                           | Number of Transactions                                                                                                                                     |
|                                                                                                                                              | CTA* (\$)        | >=1                                                                                                                 | >=1                       | 500                        | 250                         | Cumulative total Amount (\$ USD)<br>(For Issuers where Visa declines on their behalf Visa recommends higher CTA values: Current = 1,000 and previous =500) |
|                                                                                                                                              | Issuer Available | Same as above                                                                                                       |                           | Same as above              |                             | Same values apply for domestic & international                                                                                                             |
|                                                                                                                                              | STIP Available   | Same as above                                                                                                       |                           | Same as above              |                             | Same values apply for domestic & international                                                                                                             |
|                                                                                                                                              |                  |                                                                                                                     |                           |                            |                             |                                                                                                                                                            |
| <b>Device</b> <ul style="list-style-type: none"><li>• Must be <b>same</b> across wallets</li><li>• Can be set at BIN / Acct. Range</li></ul> | TTL              | The device time to live parameter is defined and set as <b>3 days less than the VIP TTL value</b>                   |                           |                            |                             |                                                                                                                                                            |
|                                                                                                                                              | NOT              | The device number of transactions parameter is defined and set as <b>5 transactions less than the VIP NOT value</b> |                           |                            |                             |                                                                                                                                                            |

**\*NOTE:** Domestic and International transactions accumulate separately for NOT and CTA settings in VIP.  
E.g. An international transaction does not add to the domestic not counter

### 3.5.2 Implementation Tasks as Token Requestor

The following sections describe the process for a token requestor to implement VTS. Issuers acting as token requestors for mobile NFC payments using HCE have the option to deploy this capability either as a standalone issuer-branded application or within their existing mobile banking application.

Issuers may also choose to leverage the services of a Token Requestor TSP to perform a portion of the activities mentioned in the following paragraphs.

#### 3.5.2.1 Develop Mobile Application Platform (MAP)

The MAP acts as an application server to the MA that runs on the cardholder's device. The MAP component is responsible for authenticating cardholders and orchestrating communication data between the VTS and Mobile Application. The communication interface between the MAP and MA is a proprietary interface defined by the issuer. The issuer is responsible for building the MAP component.

The following describes the MAP's required functionality:

- Generate the device ID to uniquely identify a device. The device ID is generated based on the device attributes.
- Generate and send device-specific certificates to encrypt and decrypt data and to sign the data to the MA.
- Generate and send device-specific symmetric key to the MA.
- Orchestrate the communication between the MA and VTS.
  - Store the data required to orchestrate the communication.
  - Generate encrypted PAN data.
  - Implement Visa-defined APIs to communicate with VTS.
- Perform customer identity verification.
- Implement business logic as per issuer requirements.
  - Required for user experience on the MA
  - Any risk decision during token and LUK provisioning

#### 3.5.2.2 Mobile Application (MA)

The issuer is required to build the MA that runs on the cardholder's device. Visa can provide an SDK to help issuers build their MA. Issuers can choose to use the Visa SDK, a third party provided SDK or implement the application on their own.

The following describes the MA's required functionality:

- User interface and experience for users to interact with the application. Visa offers consultancy to the issuer to define the user experience.
- Implement required APIs to interact with the Visa certified NFC Payment SDK (if issuer uses such SDK). For details, issuers should refer to the *Visa Digital Solutions Developers Guide*.
- Implement the logic to interact with the MAP.

---

### 3.5.2.3 Certificates

The issuer and Visa exchange certificates to establish end-to-end encryption between VTS and the Visa certified NFC Payment SDK. Visa requires an issuer parent certificate (including public key) for device signature validation and encryption. This certificate is sent out-of-band to Visa. A CBK public key is shared out-of-band with the issuer. The SDK is bundled with a corresponding CBK private key as part of the build process.

---

### 3.5.2.4 Access the Visa Digital Solutions Developer Center (Sandbox)

The sandbox environment is available for token requestors to test the platform components they develop, mainly the Mobile Application and MAP. Eligible token requestors may obtain access to the sandbox from their Visa representative. The token requestor needs to assign a developer portal administrator who then has administrative rights to make developer portal configurations and grant access to additional technical staff. With access to the sandbox the token requestor is able to:

- Generate the Visa Digital Solutions API key and shared secret
- Add additional issuer developer accounts
- Create MA profiles
- Access the Visa Provided certified NFC Payment SDK (in case client contracts to use this SDK) and development documentation

To gain access to the sandbox:

- The token requestor completes the VTS Token Requester Onboarding Form and designates a sandbox administrator. Visa assigns a Token Requester ID (TRID) to the issuer wallet.
- A Visa VTS integration specialist configures access for the issuer.
- The token requestor sandbox administrator receives a welcome email with a link to access the sandbox. After setting a password the system administrator has the ability to grant access to additional token requestor representatives as required.

For more details on using the Visa Digital Solutions Developer Center, issuers should refer to Appendix B, Visa Digital Solutions Developer Center.

### 3.5.2.5 Connecting to the Visa Digital Solutions Developer Center (Sandbox)

The following is required for a token requestor to connect to VTS APIs through the Visa Digital Solutions Developer Center:

- Access to the Visa Digital Solutions Developer Center: <https://developer.digital.visa.com/>.
- Generate an API key, which authorizes access to the VTS service.
- If the token requestor selects to use one-way SSL to connect between the MAP and VTS, then it gets a root certificate for GeoTrust to establish one-way SSL to secure a communication channel.
- If the token requestor selects to use mutual SSL to connect between the MAP and VTS, then it exchanges certificates with Visa and establishes firewall rules to secure a communication channel.

## 3.6 Production Implementation Tasks

This section describes the production implementation tasks for issuers.

### 3.6.1 Implementation Tasks as Token Issuer

#### 3.6.1.1 Complete Required Issuer Host Changes

No additional issuer host changes are required after the successful completion of issuer host testing in the sandbox and VCMS.

#### 3.6.1.2 Define Product Eligibility

Visa requires PAN BIN and PAN range information from the issuer to perform a product eligibility check for provisioning. The issuer needs to provide the following information about PAN BIN and PAN range.

- Issuer BID (Business ID)
- Product Type(s) (e.g., credit, debit, prepaid)
- PAN BIN
  - BIN Range Start
  - BIN Range End

### 3.6.1.3 Set Up Token BINs

VTs generates tokens from dedicated token BIN ranges that are mapped to PAN BIN ranges the issuer provides. Table 3–4 provides a sample of how a PAN BIN range maps to a token BIN range for token generation.

This step takes place in the sandbox but needs to be repeated in the production environment because sandbox settings do not transfer to production automatically. The issuer must configure all the supported BINs in the production environment. The process and steps in production are same as for the sandbox.

### 3.6.1.4 Issuer Host Key Management

Visa requires a MDK to create and provision tokens and to validate token transaction cryptograms during token processing. Issuers have the following options for generating MDKs for VTS:

- Visa creates new MDKs and holds them (not transmitted to the issuer). This option is recommended for issuers that do not have existing chip programs.
- Visa can use existing issuer MDKs already on file with Visa. This option is recommended for existing chip issuers.
- Issuer creates new MDKs and transmits them to Visa via an existing ZCMK.
- Visa creates new keys and transmits them to the client via an existing ZCMK.

### 3.6.1.5 Define Lifecycle Management

No changes are required to the lifecycle management settings after the successful completion of testing in the sandbox and VCMS environment.

### 3.6.1.6 Set Up Notifications and Alerts

No changes required to the notification and alerts after the successful completion of testing in the sandbox and VCMS environment.

### 3.6.1.7 Define Issuer LUK Replenishment Parameters

Issuers must set up the LUK parameters for the production environment. In the sandbox environment, Visa performs the configuration but in the production environment, the issuer is required to complete this step. The recommended values for the production system are shown in Figure 3–1, Visa LUK Thresholds Guidelines.

### 3.6.2 Implementation Tasks as Token Requestor

The following sections describe the process for a token requestor to implement VTS. Issuers acting as token requestors for mobile NFC payments using HCE have the option to deploy this capability either as a standalone issuer-branded application or within their existing mobile banking application.

#### 3.6.2.1 Access the Visa Digital Solutions Business Center (Production)

When the token requestor is ready to move to a production environment, the token requestor must migrate from the Visa Digital Solutions Developer Center (Sandbox) to the Visa Digital Solutions Business Center (Production). To migrate to the production environment:

- Visa and the token requestor review the VTS Token Requestor Onboarding Form to confirm system configuration. The issuer may choose to use the same settings as the sandbox.
- A VTS integration specialist creates a new system account and migrates all configuration settings from the sandbox.
- The token requestor's Visa Digital Solutions Business Center administrator receives a welcome email with a link to access the Visa Digital Solutions Business Center. After setting a password, the system administrator has the ability to grant access to additional token requestor representatives as required.

To migrate to the production environment the token requestor must perform the following activities:

- Change the sandbox API end points to production URIs.
- Ensure that the existing functionality was tested end to end, including payments.

For more details on using the Visa Digital Solutions Business Center, issuers should refer to Appendix B, Visa Digital Solutions Developer Center.

#### 3.6.2.2 Certificates

The issuer and Visa exchange certificates to establish end-to-end encryption between VTS and the Visa certified NFC Payment SDK.

Visa requires an issuer parent certificate (including public key) for device signature validation and encryption. This certificate is shared out-of-band to Visa. The production certificate must be different than sandbox certificate.

An IBK public key is shared out-of-band with the issuer. The SDK is bundled with the corresponding CBK private key as part of the build process. The production IBK is specific to the issuer and is not the same as the sandbox IBK.

Production confidentiality and integrity certificate ID is shared with the issuer as an out-of-band processing.

---

### 3.6.2.3 Connecting to the Visa Digital Solutions Business Center (Production)

The following is required for a token requestor to connect to VTS APIs through the Visa Digital Solutions Business Center:

- Access to the Visa Digital Solutions Business Center: <https://business.digital.visa.com/>.
- Generate an API key, which authorizes access to the VTS service.
- Get a root certificate for GeoTrust to establish one-way SSL to secure communication channel between the token requestor and VTS.

For more information, issuers should refer to the *Visa Digital Solutions Developers Guide*.

---

### 3.6.3 Customer Service

Prior to making services available to cardholders, issuers and token requestors should provide training to customer service and customer-facing staff. Customer service staff should be able to assist cardholders with queries relating to the issuer:

- Activating tokens
- Identifying and researching token transactions
- Service information
- Provisioning process
- Updating cards (lost, stolen, add, remove)

---

#### 3.6.3.1 Call Center Support

Token requestors and issuers should review and update their call center procedures and scripts to support token-specific consumer inquiries. Types of inquiries might include general service, token payment transaction, token lifecycle management, and device-specific inquiries.

## 3.7 Testing

This section describes the tests that are performed to help ensure that the issuer system and VTS work as expected when integrated.

### 3.7.1 Business Enhancement Release Testing

Issuers need to work with their Visa representative to test the issuer host changes described in the *VisaNet Business Enhancements Global Technical Letter and Implementation Guide*.

### 3.7.2 Connection Tests

Connection tests are done after the environment and flow preparation are complete. The IT teams on the issuer and VTS side help ensure that the certificates exchanged between the two entities are valid. After the IT teams have confirmed that the certificates are valid, test messages are sent to validate that the communication flow between the two endpoints is enabled.

### 3.7.3 End-to-End Tests

Issuers should test their MA and SDK integration using the sandbox environment before moving to the production environment. Testing should include:

- Notifications (web and ISO)
- Negative cases
- Transaction validation in VCMS
- Personalization validation

### 3.7.4 Mobile Application Functional Testing

Visa's MA functional compliance program is administered by the Visa Ready Program office. Functional compliance is ensured when the MA meets all requirements of the *Visa Cloud-Based Payments Contactless Specification*. When a Visa provided certified NFC Payment SDK is used, Visa help ensures compliance to these functional requirements. If the issuer or token requestor develops a proprietary implementation of the *Visa Cloud-Based Payments Contactless Specification*, the issuer or token requestor can self-test by utilizing Visa-provided test plans and/or Visa-approved test tools supplied by vendors listed on the Visa Technology Partner website's (<https://technologypartner.visa.com>) public directory of test tool vendors.

In certain Visa Europe markets, Consumer Choice mandates requires that co-badged cards provisioned into digital wallets have a clear and transparent choice between payment schemes at the time of provisioning and each transaction.

Please check with your Visa Europe representative, on Consumer Choice requirements and functional testing requirements to meet these mandates.

### 3.7.5 Mobile Application Security Testing

The MA should meet all the security requirements outlined in the *Visa Cloud-Based Payments Program Minimum Requirements and Guidelines*. Visa has developed a MA security testing program administered by the Visa Ready Program office and has engaged with independent labs that provide testing based on the security evaluation requirements described in the *Visa Cloud-Based Payments Program Minimum Requirements and Guidelines*. When a Visa provided certified NFC Payment SDK is used, Visa help ensures compliance to a majority of the security requirements that can be addressed with the SDK. Issuers and token requestors are encouraged to engage with a security lab to validate against compliance to all of Visa's MA security requirements. The Visa Technology Partner website (<https://technologypartner.visa.com>) provides a list of labs that can be used for security testing.

### 3.7.6 Solution Approval

Visa's MA certification program is administered by the Visa Ready Program office. Issuers are required to share their MA with the Visa testing team to perform the set of mandatory test cases. Issuers also can test the using the Visa-provided test plans and share test results.

The Visa Ready Program office reviews the results and provides the necessary solutions approval for the commercial launch of the issuer wallets.

## 4 Implementation Checklists

This chapter provides an overview of the tasks in each of the following phases:

- Pre-implementation phase
- Implementation phase for the sandbox
- Implementation phase for production

### 4.1 Pre-Implementation Phase Checklist for HCE Program

Table 4–1 provides a checklist for the pre-implementation phase for an HCE program.

**Table 4–1: Pre-Implementation Phase Checklist for HCE Program**

| Task No. | Task                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Responsible |
|----------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1        | Project team and kickoff.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | Visa        |
| 2        | Meet with the issuer to present VTS overview.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | Visa        |
| 3        | Help ensure issuer has reviewed key artifacts: <ul style="list-style-type: none"> <li>• Standards/specifications, Visa Token Service Description, Visa Token Service Implementation Guide for Participant-Branded Solutions (Host Card Emulation – Issuer Wallet), Visa Digital Solutions Developers Guide, SDK (if applicable), package.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Issuer      |
| 4        | Complete VTS project scoping: <ul style="list-style-type: none"> <li>• Use case analysis</li> <li>• Issuer host considerations</li> <li>• Issuer MA <ul style="list-style-type: none"> <li>– Functional and application security testing</li> <li>– CDCVM and payment mode options</li> <li>– Token provisioning / activation decisions (Cardholder ID&amp;V)</li> <li>– Integration Options (Flow 1 and Flow 2)</li> </ul> </li> <li>• Issuer to establish support operations (call center)</li> <li>• Define issuer key management options (MDKs)</li> <li>• Define Active Key Management (LUK thresholds)</li> <li>• Define lifecycle management process</li> <li>• Define desired alerts and notifications</li> <li>• Provide the issuer master certificate (lead time for installation is 7 days)</li> </ul> | Issuer      |

## Visa Token Service Implementation Guide for Participant-Branded Solutions (Host Card Emulation—Issuer Wallet): Visa Supplemental Requirements

| Task No. | Task                                                                                                          | Responsible |
|----------|---------------------------------------------------------------------------------------------------------------|-------------|
| 5        | Execute Visa Token Service Issuer Participation Agreement.                                                    | Visa/Issuer |
| 6        | Complete Visa Token Service Issuer PIF (Optional).                                                            | Issuer/Visa |
| 7        | Create implementation project plan.                                                                           | Visa/Issuer |
| 8        | Verify that the issuer has access to Visa Access.                                                             | Visa        |
| 9        | Assign new services to Visa Access – VCMM and TLCM.                                                           | Visa        |
| 10       | Create Visa Access ID(s) for new users as per issuer request and assign VCMM and TLCM access to issuer users. | Visa        |

### 4.2 Implementation Phase Checklist for HCE Program (Sandbox)

Table 4–2 provides a checklist for the sandbox implementation phase for an HCE program.

**Table 4–2: Implementation Phase Checklist for HCE Program (Sandbox)**

| Task No. | Task                                                                                                                                                                                                                | Responsible |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1        | Submit Token Requester ID request form.                                                                                                                                                                             | Issuer      |
| 2        | Submit VTS Token Requestor Onboarding Form.                                                                                                                                                                         | Issuer      |
| 3        | Complete token requestor onboarding - API/shared secret key generation.                                                                                                                                             | Visa/Issuer |
| 4        | Submit configuration information.<br>Prepare information for PAN ranges to be tokenized for each funding source (e.g., credit, debit, prepaid), Business ID (BID), and a listing of the BINs to be enrolled in VTS. | Visa        |
| 5        | Submit card metadata using VCMM.                                                                                                                                                                                    | Visa        |
| 6        | Identify which MDK will be used for VTS participation.                                                                                                                                                              | Issuer/Visa |
| 7        | Submit the wallet provider master certificates to Visa.                                                                                                                                                             | Issuer/Visa |
| 8        | Submit the master certificates to Visa.                                                                                                                                                                             | Issuer      |
| 9        | Submit the Visa certificate IDs to issuer.                                                                                                                                                                          | Visa        |

## Visa Token Service Implementation Guide for Participant-Branded Solutions (Host Card Emulation—Issuer Wallet): Visa Supplemental Requirements

| Task No. | Task                                                                                                                                                                                                                               | Responsible |
|----------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 10       | Finalize lifecycle management options: <ul style="list-style-type: none"> <li>• TLM: If chosen, request access through Visa Access</li> <li>• ISO messages</li> <li>• VAUS</li> <li>• Token requestor web services API.</li> </ul> | Issuer/Visa |
| 11       | [Optional] Select desired notifications: <ul style="list-style-type: none"> <li>• Provisioning, lifecycle events, LUK threshold alerts</li> </ul>                                                                                  | Issuer      |
| 12       | Complete issuer host changes and verify that changes comply.                                                                                                                                                                       | Visa        |
| 13       | Complete token requester API integration.                                                                                                                                                                                          | Issuer      |
| 14       | Complete MA development.                                                                                                                                                                                                           | Issuer      |
| 15       | Submit the MA to Visa for Visa functional and branding compliance.                                                                                                                                                                 | Issuer/Visa |
| 16       | Complete issuer host testing with Visa GCT.                                                                                                                                                                                        | Issuer/Visa |
| 17       | Perform end-to-end LCM event.                                                                                                                                                                                                      | Issuer/Visa |

### 4.3 Implementation Phase Checklist for HCE Program (Production)

Table 4–3 provides a checklist for the production implementation phase for an HCE program.

**Table 4–3: Implementation Phase Checklist for HCE Program (Production)**

| Task No. | Task                                                                                                                                                                                                                | Responsible |
|----------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 1        | Complete token requestor onboarding – API/shared secret key generation.                                                                                                                                             | Visa/Issuer |
| 2        | Submit configuration information.<br>Prepare information for PAN ranges to be tokenized for each funding source (e.g., credit, debit, prepaid), Business ID (BID), and a listing of the BINs to be enrolled in VTS. | Visa        |
| 3        | Submit card metadata using VCMM.                                                                                                                                                                                    | Visa        |
| 4        | Identify which MDK will be used for VTS participation.                                                                                                                                                              | Issuer/Visa |
| 5        | Submit the wallet provider master certificates to Visa.                                                                                                                                                             | Issuer/Visa |
| 6        | Submit the master certificates to Visa.                                                                                                                                                                             | Issuer      |

## Visa Token Service Implementation Guide for Participant-Branded Solutions (Host Card Emulation—Issuer Wallet): Visa Supplemental Requirements

| Task No. | Task                                                                                                                                                                                                                           | Responsible |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------|
| 7        | Submit the Visa certificate IDs to issuer.                                                                                                                                                                                     | Visa        |
| 8        | Finalize lifecycle management options: <ul style="list-style-type: none"><li>• TLCM: If chosen, request access through Visa Access</li><li>• ISO messages</li><li>• VAUS</li><li>• Token requestor web services API.</li></ul> | Issuer/Visa |
| 9        | [Optional] Select desired notifications: <ul style="list-style-type: none"><li>• Provisioning, lifecycle events, LUK threshold alerts</li></ul>                                                                                | Issuer      |
| 10       | Complete issuer host changes and verify that changes comply.                                                                                                                                                                   | Visa        |
| 11       | Submit the MA to Visa for Visa functional and branding compliance.                                                                                                                                                             | Issuer/Visa |
| 12       | Complete issuer host testing with Visa GCT.                                                                                                                                                                                    | Issuer/Visa |
| 13       | Perform end-to-end LCM event.                                                                                                                                                                                                  | Issuer/Visa |

## A Token Lifecycle Management

Token requestors have the ability to perform lifecycle management on their tokens to ensure that VTS reflects the current status of their tokens. The status of a token can change at any point during its issuance, for example, due to an update to the PAN expiry date or a fraud alert on the PAN.

Maintenance of these lifecycle management events is important and helps ensure the integrity of the payment transaction.

The management of lifecycle events can be initiated through the TLCM tool on Visa Access, or via ISO messages or web services into VTS (if applicable).

The functions of lifecycle management are as follows:

- Delete/deactivate token
- Suspend token
- Resume token
- Token inquiry
- Update token expiration
- Replace PAN mapped to token

Issuers can subscribe to use TLM tool during the VTS enrollment process to manage lifecycle events.

Table A–1 reflects lifecycle management functions and interface options available to manage lifecycle events. ISO and web service options may require additional development work.

**Table A–1: Lifecycle Management**

| Function                                 | Description                                                                                                                                                             | From            | To               | Interface Options |             |           |
|------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|-------------------|-------------|-----------|
|                                          |                                                                                                                                                                         |                 |                  | ISO               | Web Service | TLCM Tool |
| Delete/<br>Deactivate<br>Token<br>[3701] | Interface to request delete of a specific token. Terminate token and PAN mapping in Visa Token Vault and trigger notifications to issuer that token is no longer valid. | Token Requestor | Visa Token Vault | 0302              | Yes         | Yes       |
| Suspend<br>Token<br>[3702]               | Request suspension of a token. Suspend action is temporary and can be resumed. No transactions are processed or approved while in suspend mode.                         | Token Requestor | Visa Token Vault | 0302              | Yes         | Yes       |

| Function                    | Description                                                                                                                              | From            | To               | Interface Options |             |           |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------|-----------------|------------------|-------------------|-------------|-----------|
|                             |                                                                                                                                          |                 |                  | ISO               | Web Service | TLCM Tool |
| Resume Token<br>[3703]      | Request resume of a suspended token. Resume action restores token to active status and allows transactions to be processed and approved. | Token Requestor | Visa Token Vault | 0302              | Yes         | Yes       |
| Token Inquiry               | Request to get all token-to-PAN mapping for a given PAN or for a specific token.                                                         | Token Requestor | Visa Token Vault | 0302              | Yes         | Yes       |
| Update Token Expiration     | Request to update token expiry date to a new date based on changes to PAN expiry date.                                                   | Token Requestor | Visa Token Vault | 0302              | Yes         | Yes       |
| Replace PAN mapped to token | Request to replace the original PAN to the token PAN mapping.                                                                            | Token Requestor | Visa Token Vault | 0302              | Yes         | Yes       |

## A.1 Visa Account Updater Suite

The VAUS is a suite of solutions that provide e-commerce and card-on-file (COF) credential updates across PANs and network tokens. There are three optional services within VAUS—Visa Account Updater (VAU), Real Time Visa Account Updater, and Visa Digital Credential Updater (VDCU).

Visa Digital Credential Updater is a value-added service that complements VTS by helping to ensure token credentials are updated when the underlying PAN or token status changes. These events include:

- Account number changes
- Expiration date changes
- Token status: suspend, delete and resume

Please note there is no impact to issuer implementation of lifecycle management.

## B Visa Digital Solutions Developer Center

The Visa Digital Solutions Developer Center (Sandbox) and Visa Digital Solutions Business Center (Production) are designed to assist the issuer with some common needs in an integration test environment and production environment. This tool gives issuers the ability to:

- Keep company information up to date
- Manage token requestor profiles
- Manage integration API keys
- Manage users to control who has access to the issuer's Visa Digital Solutions account and information

